Allergen Detection and Allergen Association by Classification

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***Abstract*—After looking at a nutrition and allergen database, we performed 6 different classification algorithms to label a food as having an allergen. Our primary metric was recall, as we were concerned with labeling a food with an allergen correctly to protect people with allergies. The best model we found was the XGBoost model for this purpose. Additionally, we fit models per allergen type (e.g. gluten) and determined features that affected the classification.**

***Index Terms*—Classification algorithms, Machine learning, Medical Conditions**

I. Introduction

OVER 80 million people have been diagnosed with allergies in the United States, and healthcare costs due to allergies exceed $18 billion annually [1]. Allergic reactions can range from mild discomfort to life-threatening conditions such as anaphylaxis, which can require immediate medical attention and, in severe cases, emergency room visits. As a result, there is a critical need for reliable and proactive methods to identify potential allergens in food products before consumption. One of the authors has a child who has needed an emergency room visit of this type.

In recent years, the growing availability of nutritional and ingredient-level data has made it possible to apply data-driven techniques to solve this problem. Machine learning models, in particular, are a promising tool for predicting whether a food product is likely to contain a specific allergen based on its composition and labeling characteristics. Beyond prediction, these models can also provide insight into the most influential features associated with different allergen categories, improving transparency and interpretability.

By identifying both high-risk products and the underlying contributing factors, this work aims to support more informed decision-making for consumers, manufacturers, and healthcare professionals. Ultimately, the goal is to reduce the likelihood of accidental allergen exposure and enhance the safety and well-being of individuals and families affected by food allergies. We hypothesize that an ensemble-based method will produce the best recalls when identifying allergens.

II. Materials and Methods

The data we are using for our analysis consists of the dataset available on Kaggle titled Global Food & Nutrition Database 2026 [2]. The initial dataframe has 4997 samples and 24 features.

Our classification pipeline began with comprehensive exploratory data analysis (EDA) to better understand the structure and nature of the dataset. We first examined the distribution of missing values across all features and computed five-number summaries for each continuous variable to analyze their distributions and detect potential anomalies. During this process, we identified an extreme outlier in the *fiber_100g* feature, with a value on the order of trillions. Given the context of nutritional data, this value was deemed implausible and likely the result of a data entry error; therefore, the corresponding sample was removed from the dataset. After removing this sample, our classification metrics improved.

To further improve data quality and model validity, we eliminated features that could introduce data leakage, contained identification-based information, or exhibited excessively high cardinality that would not contribute meaningful predictive signal. Here are the correlations between the continuous columns:

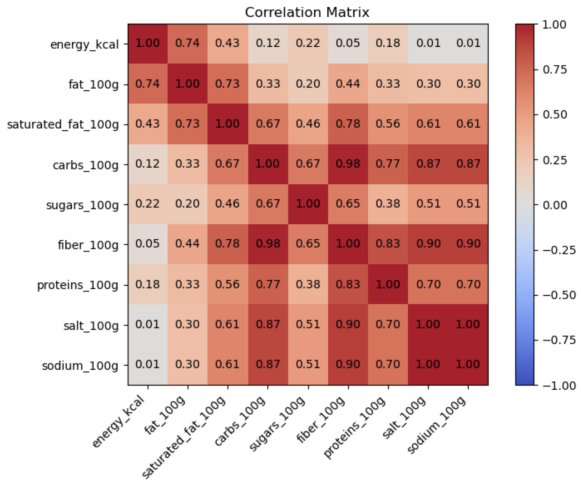

Fig. 1. Correlations among continuous columns

One insight from the correlation analysis in Figure 1 is that there is a perfect linear relationship between *sodium_100g* and *salt_100g*. This is expected due to the fact that table salt contains a proportional amount of sodium. To avoid redundancy and potential multicollinearity issues, the *sodium_100g* feature was excluded. The metrics (recall and precision) improved after dropping the perfectly correlated column. One can see from Figure 1 that there are no negative correlations and several of the other correlations are quite high, such as between fiber and carbs with 0.98.

We also investigated the distribution of each of the continuous columns. Here are histograms for the continuous columns (figure 2):

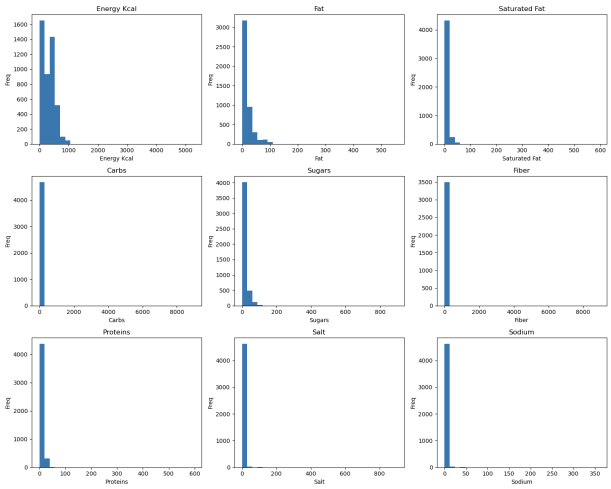

Fig. 2. Histograms of continuous columns

From looking at these histograms in Figure 2, we believe that the distributions were extremely skewed. We calculated skewness metrics and we present them in the following table:

TABLE I
SKEWNESS OF NUTRITIONAL FEATURES

Feature	Skewness
Energy (kcal)	2.758
Fat (100g)	4.709
Sugars (100g)	15.054
Saturated Fat (100g)	24.231
Proteins (100g)	32.253
Salt (100g)	49.537
Sodium (100g)	49.538
Fiber (100g)	58.731
Carbohydrates (100g)	64.027

It is clear from the skewness metrics (where any number over 1 indicates high skewness) in Table I that all of the distributions are highly right-skewed.

After completing these preprocessing steps, the final dataset for classification consisted of 4,996 samples and 12 features. The training and validation set together had 3996 samples. The training set alone had 3196 samples, and the validation set had 800 samples. Finally, the test set had 1000 samples. All feature matrices had 11 features.

Using this refined dataset, we evaluated six classification algorithms: Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), Regularized LDA, XGBoost, LightGBM, and TabNet. These models were selected to provide a comparison across linear, nonlinear, and ensemble-based approaches. Model performance was primarily evaluated using recall, with precision serving as a secondary metric. Emphasizing recall was critical in this application, as the cost of failing to identify a food product containing an allergen (false negatives) is significantly higher than incorrectly flagging a safe product (false positives). This prioritization aligns with the overarching goal of minimizing the risk of harmful allergen exposure.

To the end of minimizing false negatives, we also iteratively adjusted the positive label threshold to maximize recall. We ensured that a minimum precision was maintained (0.7), and the recall obtained from this optimization was used on the held-out test set in order to rank the models.

In addition to classification, we also performed association analysis between the presence of different allergens and certain nutrients in food. This analysis was conducted in a separate notebook and started with dropping identification-based columns such as 'brands', 'product_name' and 'allergens'. We then identified the six 'contains' columns indicating the presence of allergens (gluten, dairy, nuts, soy, eggs, fish).

Our association pipeline began with preprocessing for both numeric and non-numeric variables. For numeric columns, we imputed missing values using K-nearest neighbors with $k = 5$ and used standard scaling. For categorical variables, we used a simple imputer with the 'most_frequent' strategy for missing values and one-hot encoding. To conduct the final association analysis, we used a linear discriminant analysis (LDA) model.

We defined a series of functions to simplify the analysis of all six allergens. For each allergen column, we began by setting that column as the target. Initially, we dropped the other five "contains_allergen" columns, but found that our results were stronger if we left them in. Thus, we later omitted the removal step and only removed the target column. We then performed identical train-test-validation split stratified by the response column to ensure equal representation of both categories (yes or no) in all sets.

We then fit the pipeline on the training set and predicted probabilities for the validation set. Then, we defined different thresholds (ranging from 0.05 to 0.95 with an increment of 0.05) and get the precision and recall scores for each threshold. The threshold of the maximum recall score becomes the threshold for evaluating on the test set.

Finally, we evaluated on the test set, obtaining final precision and recall scores plus a confusion matrix. We also obtained a dataframe of the model coefficients. This process was performed on all six allergen variables present in the dataset.

III. RESULTS

To begin classification, we calculated the baseline recall by computing the mean value of the response (has allergies) in the test set. We obtained a value of 0.622.

After training the classification algorithms on the training data and tuning them on the validation set, we evaluated them on the testing set. Here are the results from the final evaluation:

TABLE II
MODEL PERFORMANCE COMPARISON

Model	Recall	Precision	Threshold
LightGBM	0.9936	0.7136	0.05
XGBoost	0.9871	0.7318	0.10
QDA	0.9711	0.7123	0.65
LDA	0.9518	0.7291	0.45
Reg LDA	0.9469	0.7037	0.45
TabNet	0.9196	0.7699	0.45

First of all, we noticed from Table II that all models were performing better than random chance (a recall of 0.622). All of the recalls we obtained were also above 0.9000. The recalls were tuned by adjusting the classification thresholds

to maximize the recall. This resulted in lower precisions, a tradeoff we believe was worth it. We decided that XGboost was the best performing from these models, as it has the second highest recall and the second highest precision. The threshold for categorizing the result as a positive was 0.10 for that model. Here is the confusion matrix for that model:

TABLE III
CONFUSION MATRIX FOR ALLERGEN CLASSIFICATION

	Predicted: No Allergen	**Predicted: Allergen**
Actual: No Allergen	153 (TN)	225 (FP)
Actual: Allergen	8 (FN)	614 (TP)

One can see from the confusion matrix in Table III that there were only 8 false negatives, which is a promising amount for our use case. However, one can also see that there were 225 false positives. Our model is cautious, but we believe that safety is a high priority when determining the presence or absence of allergens. Here are the feature importances for the XGBoost model:

TABLE IV
TOP FEATURES RANKED BY IMPORTANCE

Rank	**Feature**	**Importance**
1	`proteins_100g`	0.1313
2	`fat_100g`	0.0532
3	`sugars_100g`	0.0452
4	`energy_kcal`	0.0447
5	`carbs_100g`	0.0447
6	`fiber_100g`	0.0371
7	`salt_100g`	0.0325
8	`saturated_fat_100g`	0.0313
9	`ecoscore_grade`	0.0261
10	`nova_group`	0.0032
11	`nutriscore_grade`	0.0029

In Table IV, we observed that protein, fat, sugar, caloric, and carbohydrate nutrition scores were the most important for the classification with the XGBoost model. Of the features mentioned, protein was the most important feature by a factor of over 2.

After running association analysis on each of the six allergen variables and observed the following top three features (in terms of absolute correlation, positive or negative) for each variable:

TABLE V
TOP THREE FEATURES FOR EACH ALLERGEN

Allergen	**Top Three Features**
Gluten	carbs_100g, contains_eggs, fat_100g
Dairy	ecoscore_grade_NOT-APPLICABLE, saturated_fat_100g, contains_eggs
Nuts	sodium_100g, salt_100g, nova_group_2.0
Soy	nova_group_2.0, contains_gluten, carbs_100g
Eggs	contains_fish, contains_gluten, fat_100g
Fish	ecoscore_grade_F, ecoscore_grade_E, contains_eggs

In Table V, we noticed that some of these associations are intuitive. For example, sodium is associated with the presence of nuts, saturated fat and eggs are associated with dairy, and carbohydrates are associated with gluten. Interestingly, many of the main predictors for certain allergens were other allergens, strengthening the validity of our decision not to drop these columns as features.

For gluten (Table VI), we obtained baseline precision and recall scores of 0.32. After running the analysis, the precision score for gluten increased to 0.533 and the recall score increased to 0.95:

TABLE VI
GLUTEN ASSOCIATION ANALYSIS RESULTS

	Precision	**Recall**
Baseline	0.32	0.32
LDA Prediction	0.533	0.95

For dairy (Table VII), the baseline value for precision and recall was 0.299. After running the LDA algorithm, we obtained a precision score of 0.543 and a recall score of 0.752 (see Table VII).

TABLE VII
DAIRY ASSOCIATION ANALYSIS RESULTS

	Precision	**Recall**
Baseline	0.299	0.299
LDA Prediction	0.543	0.752

For nuts (Table VIII), we obtained initial scores of 0.118 for precision and recall. The precision score increased to 0.584 and the recall score increased to 0.56 (see Table VIII).

TABLE VIII
NUTS ASSOCIATION ANALYSIS RESULTS

	Precision	**Recall**
Baseline	0.118	0.188
LDA Prediction	0.584	0.56

For soy (Table IX), we obtained initial precision and recall scores of 0.17. From Table IX, we can see that the precision score for soy increased to 0.559 and the recall score increased to 0.426:

TABLE IX
SOY ASSOCIATION ANALYSIS RESULTS

	Precision	**Recall**
Baseline	0.17	0.17
LDA Prediction	0.599	0.426

For eggs (Table X), we obtained a baseline precision/recall score of 0.064. After running association analysis, the precision score increased to 0.428 while the recall score increased to 0.187. This can be seen in Table X:

TABLE X
EGGS ASSOCIATION ANALYSIS RESULTS

	Precision	**Recall**
Baseline	0.064	0.064
LDA Prediction	0.428	0.187

Finally, for fish (Table XI), we obtained baseline scores of 0.022 for precision and recall. The precision score increased to 0.333 and the recall score increased to 0.54 (see Table XI).

TABLE XI
FISH ASSOCIATION ANALYSIS RESULTS

	Precision	Recall
Baseline	0.022	0.022
LDA Prediction	0.333	0.54

We observe that our association analysis model has very low precision and recall for every variable except gluten. This could be partly due to a lack of samples containing the other allergens. We intend to tune our model hyperparameters to obtain sounder results since we are especially concerned about low recall because false negatives are more problematic than false positives when determining whether or not something contains an allergen. However, we observe that all precision and recall scores after running the analysis are much higher than the baseline scores, indicating that even the weakest model performed better than random chance. The model is at least somewhat effective at predicting the presence of allergens, but further tuning is necessary.

IV. DISCUSSION AND SUMMARY

Our project aimed to take nutritional features of a dataset of foods and use machine learning algorithms to accurately predict whether a food item had an allergen of any kind. The best algorithm according to the primary metric of recall and secondary metric of precision was XGBoost with a recall of 0.9871 and a precision of .7318. Protein, fat, and sugar nutritional content were the top three features for this prediction. Our next task was to separately analyze the feature importances for the different allergens and find associations that impacted the prediction for the presence of the allergen. Some of the insights were very reasonable, such as carbohydrate nutrition making gluten more likely to be present or high sodium making nuts more likely to be present.

One challenge we faced during this research was a lack of samples containing certain allergens (eggs, fish), which likely had a negative impact on our association analysis results. By obtaining additional samples containing these allergens, we may be able to achieve more robust results. Additionally, one potential future challenge involves our use of nutrient-level data rather than ingredient-level data. Some practical applications might require the use of ingredient-level analysis whereas our models have been trained on nutrients. This may make real-world deployment challenging and might require the introduction of ingredients as other predictors of allergens in addition to nutrient-level information.

Future work could involve using more models on the initial machine learning such as random forests and on improving the classification metrics for the separate allergen models, as some of them performed quite poorly. Additionally, eventual deployment of our best model to a personal nutrition application may be beneficial to individuals with food allergies. We envision an app designed to be used while shopping for groceries that could alert individuals to the potential presence of allergens that may not be well-documented on food labels. Our overall vision for this project is to address the need for reliable allergen identification methods in real-world situations.

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SELF-DECLARATION OF WORK

Aaron Schneberger: I completed the EDA notebook and the classification notebook in the repository; in the report, I completed the introduction and contributed to the ”Materials and Methods”, ”Results”, and ”Discussion and Summary” sections.

Noelle Haviland: I completed the association analysis, contributed to organization of work in our repository, and contributed to the ”Materials and Methods”, ”Results”, and ”Discussion and Summary” sections of the report.

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